Namhoon Kim

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Education

Georgia Institute of Technology Aug 2024 - present
Ph.D. Student in Machine Learning

- **Advisor**: Sara Fridovich-Keil
- **Research Interest**: Computational Imaging, 3D Reconstruction, Uncertainty Quantification

Yonsei University Mar 2019 - Aug 2024
B.S. in Electrical and Electronic Engineering

- **GPA**: 3.94/4.3, *Magna Cum Laude (Top 3 % in Engineering School)*, **Major GPA**: 4.13/4.3
- **Senior Thesis** (Advisor: Seung Ah Lee) End-to-End, Physics-Informed Lensless Segmentation Network.

Publications

Conferences

- **N. Kim** and S. Fridovich-Keil, "Task-Aligned Implicit Neural Representations" *CVPR*, 2027 (under review).
- **N. Kim**, D. Bao, S. Jeong, and S. Fridovich-Keil, "CARE-INR: CArrier REsidual Implicit Neural Representations" *CVPR*, 2027 (under review).
- **N. Kim***, N. Moeini*, J. Romberg, and S. Fridovich-Keil, "3D Field of Junctions: A Noise-Robust, Training-Free Structural Prior for Volumetric Inverse Problems" *ECCV*, 2026.
- **N. Kim** and S. Fridovich-Keil, "Grids Often Outperform Implicit Neural Representation at Compressing Dense Signals" *NeurIPS*, 2025.
- **N. Kim** and S. Fridovich-Keil, "Towards Distribution-Shift Uncertainty Estimation for Inverse Problems with Generative Priors" *IEEE CAMSAP* (invited paper), 2025.
- K. Lee, J. Bae, **N. Kim**, N. Beak, T. Kim, W. Park and S. Lee, "Differentiable Lensless Camera Design with Extended Field of View" *Photonics Technology (at Optics and Photonics Congress)*, 2023.
- N. Baek, K. Lee, J. Bae, J. Park, T. Kim, D. Bae, **N. Kim**, W. Park and S. Lee, "Single-shot fabrication of light modulating phase mask for lensless camera using off-the-shelf components" *Photonics Technology (at Optics and Photonics Congress)*, 2023.

Journals

- S. Jeong, **N. Kim**, and S. Fridovich-Keil, "Revisiting Shearlet Thresholding for 3D Denoising" *IEEE Transactions of Computational Imaging (TCI)*, 2026 (under review).
- G. Jeong, S. Cho, M. Shon, **N. Kim**, P. Venkatesan, W. Hwang, K. Seo, S. Lim, W. Kim, D. Ha, R. Ranade, K. Youn, R. Cherukuri, Y. Wang, S. Datta, A. Khan, and S. Yu "Physics-Informed Neural Operator Surrogate for FeNAND Retention Prediction" *IEEE Journal of the Electron Devices Society (J-EDS)*, 2026 (under review).
- S. Lee, **N. Kim**, and S. Yoo, "Design-Space Analysis of Plug-and-Play Diffusion for PDE-Constrained Inverse Problems" *TMLR*, 2026 (under review).
- **N. Kim**, A. Pananjady, A. Pourmorteza and S. Fridovich-Keil, "Perfusion Imaging and Single Material Reconstruction in Polychromatic Photon Counting CT" *Medical Physics*, 2025 (under review).

Workshops

- **N. Kim***, N. Moeini*, J. Romberg, and S. Fridovich-Keil, "3D Field of Junctions: A Noise-Robust, Training-Free Structural Prior for Low-Dose CT" *4th workshop on Vision-based InduStrial InspectiON (VISION) (at CVPR)*, 2026.
- **N. Kim** and S. Fridovich-Keil, "Uncertainty Quantification for Inverse Problems with Generative Priors under Distribution Shift" *Statistical Frontiers in LLMs and Foundation Models (at NeurIPS)*, 2024.

Selected Honors and Awards

Next-generation Science and Technology Leader NET, 2025
Korean Federation of Science and Technology Societies
- Awarded $2,000 research grant from KOFST supported interdisciplinary research projects.
  "Efficient Management of Data Centers for Extreme Environments using Digital Twin Technology"

2nd prize, ML@GT Student Conference Best Poster Award: Awarded $200 Sep 2025

2nd prize, Mobility SW Hackathon, Hyundai Mobis Feb 2023
- Developed an In-vehicle Noise Resistant Voice Recognition model with deep neural network.
- Awarded $4,000 for the prize and registered as an excellent talented human resource in Hyundai.

3rd prize, Yonsei Dream Library: Metaverse Content Production Contest Feb 2022
- Awarded $800 for prize for developing a metaverse virtual university library using Lua and Roblox IDE.
- Established a server-client communication enabling the construction of a competitive studying system based on the exchange of data between users.

Mentoring

Sungwon Jeong, M.S. student at Georgia Institute of Technology Fall 2025 - present

Scholarships

Yonsei Eminence Scholarship, Yonsei University Fall 2022 - Spring 2024
- Awarded for academic excellence:
  $3000 (2/3 of Tuition) at Spring 2023, Fall 2023; $2000 at Spring 2024; $1000 at Fall 2022

Teaching Experiences

Peer Tutor of course Digital Signal Processing (EEE4420), *Yonsei University* Winter 2024
- Provided 16 hours of tutoring for 10 undergraduate students

Residential College Program: Arduino Programming & Operation, *Yonsei University* Spring 2023
- Appointed as a tutor and assisted 30 hours of lecture for 60 freshmen

Social Engagement: Yon-In High School Students Mentoring, *Yonsei University* Fall 2019
- Mentored high school students during the semester

Peer Tutor of course Electricity & Magnetism(1) (EEE2030), *Yonsei University* Summer 2022
- Provided 16 hours of tutoring for 40 undergraduate students

Leadership and Military Service

Republic of Korea Defense Intelligence Command (Military Service) Jul 2020 – Jan 2022
Army Sergeant, honorable discharge

YAI, Artificial Intelligence Club *Yonsei University — President* Dec 2023 – Jun 2024

Roboin, Robotics Club *Yonsei University — Executive Staff* Mar 2023 – Aug 2023

Jamsin High School — *Student Vice President* 2014

Skills

Programming: Python, MATLAB, C/C++, C#, Lua, Verilog

Software and Tools: PyTorch, Linux, HTML/JS/CSS, Latex, Unity, CAD (Blender, Fusion 360, PTC Creo)

Experiment: Optical systems setup, Arduino, Raspberry Pi, 3D Printing (Form3+, Ender)

Soft Skills: Time management, Leadership, Teamwork